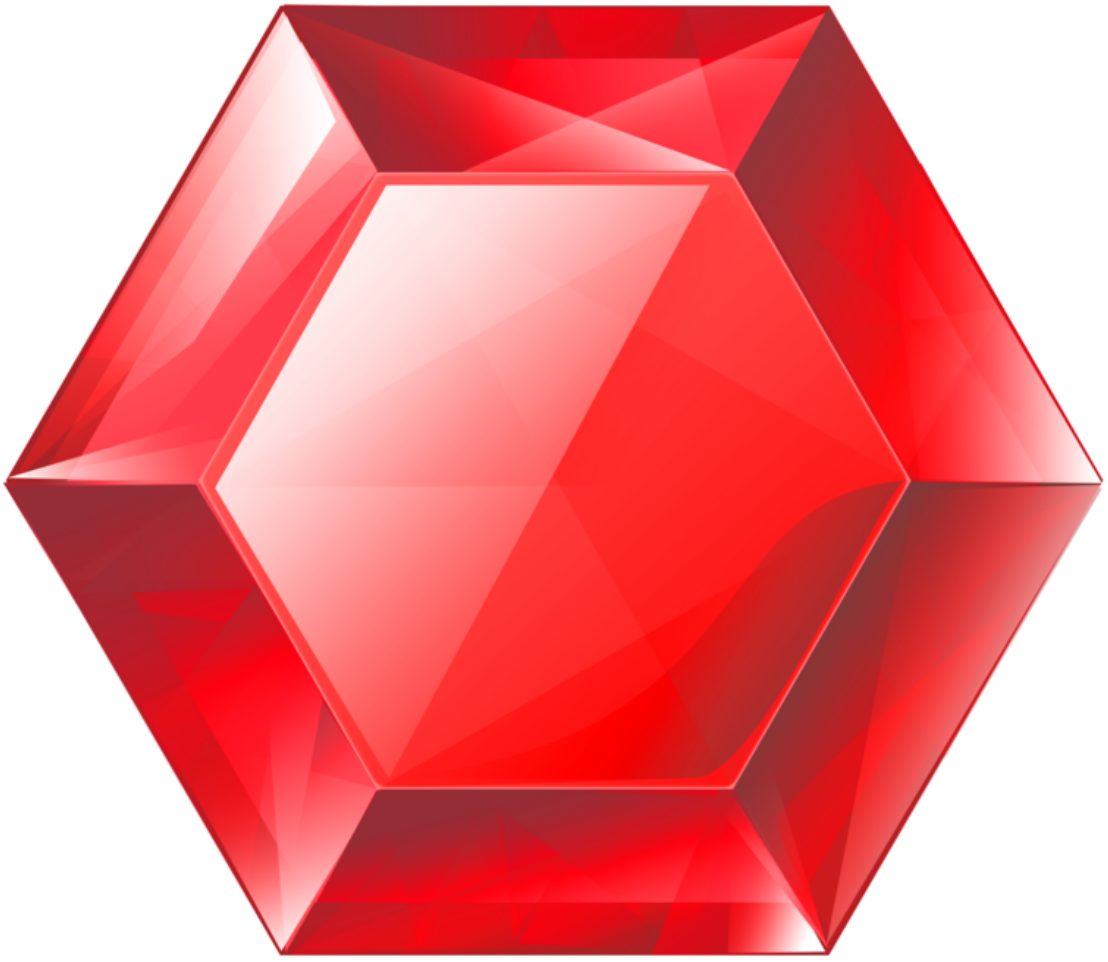
gemstone-js vs gemstone-py

TypeScript and Python GemStone/S bridge comparison



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gemstone-js vs gemstone-py

`gemstone-py` and `gemstone-js` solve the same broad problem from different runtime ecosystems. `gemstone-py` is the more mature Python path. `gemstone-js` is the async TypeScript path for JavaScript services, CLI tools, and npm-based delivery.

Use this guide when choosing between Python and TypeScript for GemStone/S application code. If the application already lives in Python, use `gemstone-py`. If the application already lives in Node, Deno, Bun, Express, Fastify, Fetch API, or Hono, `gemstone-js` is the natural fit once its native runtime path is proven for your platform.

Quick Answer

Question	Better default
Production Python app or script	`gemstone-py`
FastAPI, Litestar, or Django service	`gemstone-py`
TypeScript service or CLI	`gemstone-js`
Express, Fastify, Fetch API, or Hono service	`gemstone-js`
Most mature visual database explorer today	`gemstone-py`
npm package and TypeScript type workflow	`gemstone-js`
Lowest operational risk today	`gemstone-py`

Best Fit

Use case	gemstone-py	gemstone-js
Scripting	Pythonic, low-friction scripts and notebooks	Possible, but async JavaScript is less convenient for small one-off scripts
Web apps	FastAPI, Litestar, Django examples	Express, Fastify, Fetch API, Hono adapters
Async model	Python sync and async APIs	Async-first API throughout

Use case	gemstone-py	gemstone-js
Package ecosystem	PyPI/TestPyPI and Python packaging	npm package shape and optional `@gemstone-js/native`
Type system	Python type hints and generated access helpers	TypeScript signatures, manifests, decorators, and generated wrappers
Runtime support	Python interpreters plus optional native acceleration	Node native package, Deno/Bun FFI starters, mock runtime
Visual tooling	Python database explorer and VS Code workbench are more mature	Doctor, inspect, examples, benchmarks, migrations, but no equivalent visual explorer yet

Maturity Matrix

Capability	gemstone-py	gemstone-js
Install path	Mature PyPI path	Alpha npm package shape, `0.1.0-alpha.0` locally
Native bridge	Optional native acceleration already integrated into the Python release lane	Optional `@gemstone-js/native`, plus Deno/Bun FFI starters
Session API	Mature sync and async session APIs	Async-first `Session.connect()`, `execute()`, `perform()`, `performWith()`
Pooling	Python web/session patterns	Session pool with reset-aware release, warmup, health checks, and lifecycle events
Request scope	FastAPI/Litestar/Django examples	Shared `RequestScope` / `TransactionScope` across Express/Fastify/Fetch/Hono
Persistent roots	Supported	`PersistentRoot`, `GsDict`, `OrderedCollection`, `GStore`
Migrations	Supported in Python docs/examples	Module-style migrations CLI and metadata roots
Codegen	Python codegen and typed access demos	Manifest/decorator TypeScript codegen with schema validation
Benchmarks		

Capability	gemstone-py	gemstone-js
	Python performance docs and native checks	Benchmark report/baseline/compare/ register tooling
Doctor/setup	Python setup docs and verification scripts	`gemstone-js-doctor` for runtime, env, native import, and optional live checks
Explorer/ workbench	Stronger today	Not yet equivalent
Live CI confidence	Broader today	Needs more platform/runtime live coverage

Where gemstone-js Is Strong

- It is async-first, which matches modern JavaScript services.
- It has framework adapters for Express, Fastify, Fetch API, and Hono.
- It has a mock runtime, useful for TypeScript unit tests without a live stone.
- It has a rich package-tooling surface: doctor, examples, inspect, migrations, benchmarks, checksums, and API-contract checks.
- Its codegen model can use manifests, decorators, schemas, generated signatures, and TypeScript compile checks.
- It has high-level JavaScript value marshalling through `performWith()` and managed handles.

Where gemstone-py Is Still Ahead

- `gemstone-py` is more mature as a published, documented user path.
- The Python database explorer and VS Code workbench are richer product surfaces today.
- Python examples cover more end-to-end workflows with lower setup friction.
- The Python release lane has more established PyPI/TestPyPI/native wheel/VSIX verification history.
- Live GemStone coverage is broader around sync, async, framework, native, and lifetime behavior.

Recommended Work Batches for gemstone-js

Batch	Work	Estimate
1	Native publish confidence: npm package, `@gemstone-js/native`, clean install, live smoke	6-10 hours
2	Visual tooling: explorer/workbench for inspect, browse, codegen, and persistent roots	10-18 hours
3	Installed examples: quickstart, web adapters, migrations, codegen, persistence helpers	5-8 hours
4	Live CI: Node/Deno/Bun, framework adapters, pools, transactions, migrations	8-14 hours
5	Documentation/release polish: Medium article, PDF docs, release checklist, checksums	5-8 hours
6	Cross-project alignment: shared concepts with gemstone-py and gemstone-rs explorers/codegen	8-14 hours

Total: roughly **42-72 hours** to bring `gemstone-js` much closer to `gemstone-py` in maturity and product polish.

CLI Comparison

From the `gemstone-rs` checkout:

```
gemstone-rs compare gemstone-js
gemstone-rs compare gemstone-js --gaps
gemstone-rs compare gemstone-js --next
gemstone-rs compare gemstone-js --batches
gemstone-rs compare gemstone-js --json
gemstone-rs compare gemstone-js --gaps --json
gemstone-rs compare gemstone-js --next --json
gemstone-rs compare gemstone-js --batches --json
gemstone-rs compare all --next
gemstone-rs compare all --next --json
```

Use the JSON form when an editor, release script, or dashboard needs to render the comparison. Use `compare all` when you want the TypeScript/Python and Rust/Python tracks in one report.

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